

Yunxiao LIU

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EDUCATION

Technische Universität München (TUM)

Munich, Germany

M.Sc. Applied and Engineering Physics

2023 – Oct. 2026

- GPA: 1.518 | Theoretical Solid State Physics (1.0), Computational Methods in Many-Body Physics (1.7)
- Thesis (in progress): Structural and magnetic properties in halogen-functionalized hybrid organic-inorganic compounds ($4X\text{-BA}$)₂MnCl₄ ($X = \text{H, Cl, I, Br}$)

University of Macau

Macau, China

B.Sc. Applied Physics and Chemistry

2019 – 2023

- GPA: 3.84/4 (equiv. 1.1 in German grading scale) | Dean's Honour List (2019–2023)
- ICBC Macau Scholarship (2021/2022); Frank Wong Scholarship (2019/2020)
- Thesis: Two-dimensional metal oxides: a first-principles study

RESEARCH EXPERIENCE

Working Student

Apr. 2024 – Sep. 2025

Forschungs-Neutronenquelle Heinz Maier-Leibnitz (FRM II)

Munich, Germany

- Investigated magnetic phase transitions of layered hybrid perovskite materials (4X-Mn-BA series, $X = \text{Cl, Br, I, H, F}$) under Dr. Jitae Park and Dr. Lukas Beddrich
- Characterised magnetic and structural properties using polarized microscope, Quantum Design PPMS, and X-ray diffraction
- Developed Python library to automate processing and visualization of PPMS magnetometry data

Undergraduate Research Assistant

Oct. 2021 – Jun. 2023

Institute of Applied Physics and Materials Engineering

Macau, China

- Synthesized novel water-splitting photocatalysts via hydrothermal methods, screening candidate materials for optimal catalytic performance
- Characterized materials using SEM, TEM, and electrochemical workstation to evaluate morphology, structure, and catalytic activity
- Performed first-principles calculations (VASP/DFT) to model electronic properties and thermodynamic stability of candidate materials

PROJECTS

PPMS_ToolKit | Python, PySide6, SQLite, Parquet, matplotlib

2024 – present

- Developed a pip-installable library for managing and visualizing Quantum Design PPMS data (VSM, Heat Capacity)
- Stores imported .dat files in SQLite with Parquet-backed caching for fast repeated loading and deduplication
- Ships a Qt6 GUI with customizable plot legends, curve toggling, and automatic ZFC/FC mode detection from filenames

vsm_visualizer | Python, DearPyGui

2025

- Lightweight desktop viewer for PPMS VSM data: split-panel UI with a file browser and live matplotlib preview, distributed as a standalone .exe requiring no installation

Recipe-app | FastAPI, SQLite, JavaScript, Docker, Mistral

2025

- Full-stack household recipe manager with ingredient inventory tracking, expiration alerts, and per-household data isolation
- Integrated OpenAI-compatible LLM API (Mistral) for AI-powered meal suggestions based on available ingredients

Personal Homelab | Raspberry Pi 5, Docker, Cloudflare Tunnel, Linux

2025 – present

- Self-hosted productivity services (Seafile, Immich, Paperless-ngx) on Raspberry Pi 5
- Exposed services securely via Cloudflare Tunnel without opening inbound firewall ports

TECHNICAL SKILLS

Languages: English (C1), German (B1), Chinese (Native)

Scientific Computing: Python, LaTeX, matplotlib, pandas, NumPy

Experimental Techniques: Quantum Design PPMS, Powder XRD, polarized light microscopy